

Unemployment Analysis with Python

A statistical study of unemployment trends in India and the impact of the COVID-19 pandemic

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Overview

This report analyzes two unemployment datasets covering Indian states: one spanning May 2019 to June 2020 (740 monthly state-level records across 28 states/UTs), and a second spanning January to October 2020 with geographic zone information. The goal is to identify trends over time, compare rural versus urban unemployment, and quantify how the COVID-19 lockdown affected the labor market.

National Unemployment Trend

The chart below shows the national average monthly unemployment rate. It held steady around 9-10% through 2019 and early 2020, then spiked sharply following the COVID-19 lockdown that began in late March 2020, peaking near 24.9% in May 2020 before beginning to recover.

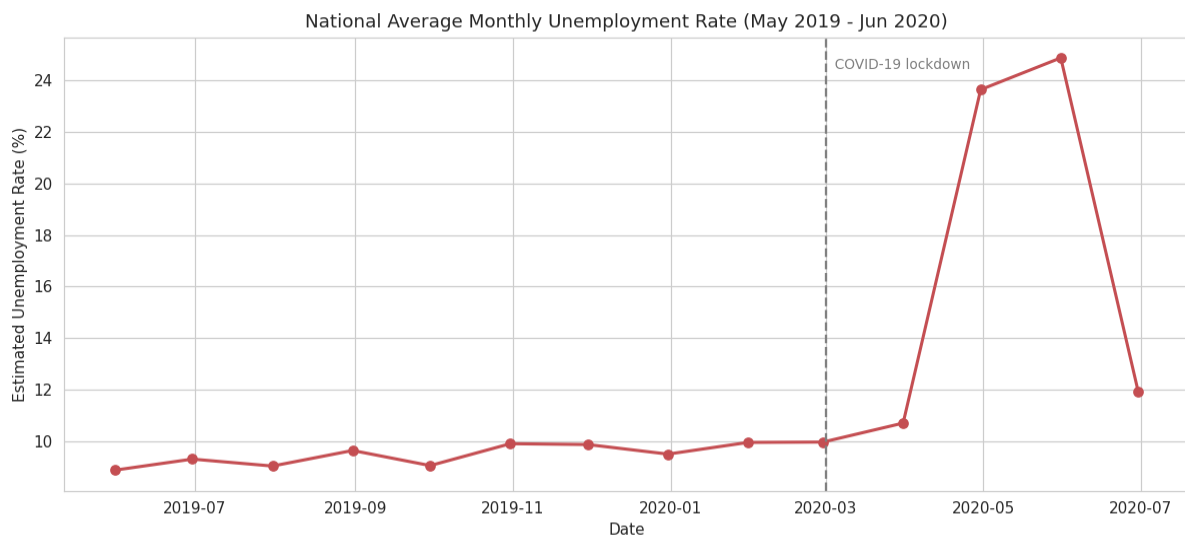


Figure 1. National average monthly unemployment rate, May 2019 - June 2020.

COVID-19 Impact Analysis

Splitting the data into a Pre-COVID period (before March 2020) and a COVID period (March 2020 onward), the national average unemployment rate rose from 9.51% to 17.77% — an increase of 8.26 percentage points. Average labour participation fell from 43.89% to 39.33% over the same split, indicating that some people also left the workforce entirely rather than remaining unemployed job-seekers.

Most-affected states	Change (pp)	Least-affected / improved	Change (pp)
Puducherry	+37.36	Assam	+0.21
Tamil Nadu	+22.57	Chandigarh	-2.00
Jharkhand	+22.07	Himachal Pradesh	-2.06

Bihar	+17.80	Tripura	-2.31
Karnataka	+12.05	Jammu & Kashmir	-4.33

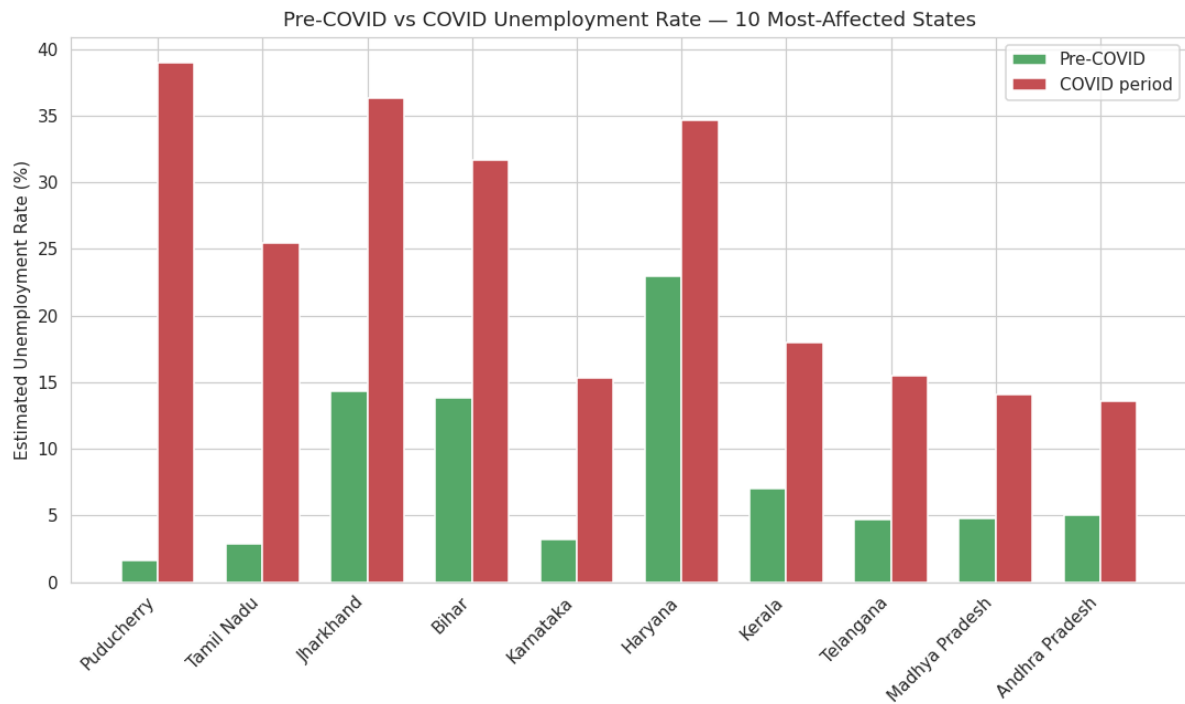


Figure 2. Pre-COVID vs. COVID-period unemployment rate for the 10 most-affected states.

State and Regional Patterns

Averaged across the full May 2019 - June 2020 window, unemployment rates vary widely by state. Using the January-October 2020 dataset's zone classification, the North zone had the highest average unemployment rate (15.89%), followed by East (13.92%) and Northeast (10.95%); the West zone had the lowest (8.24%).

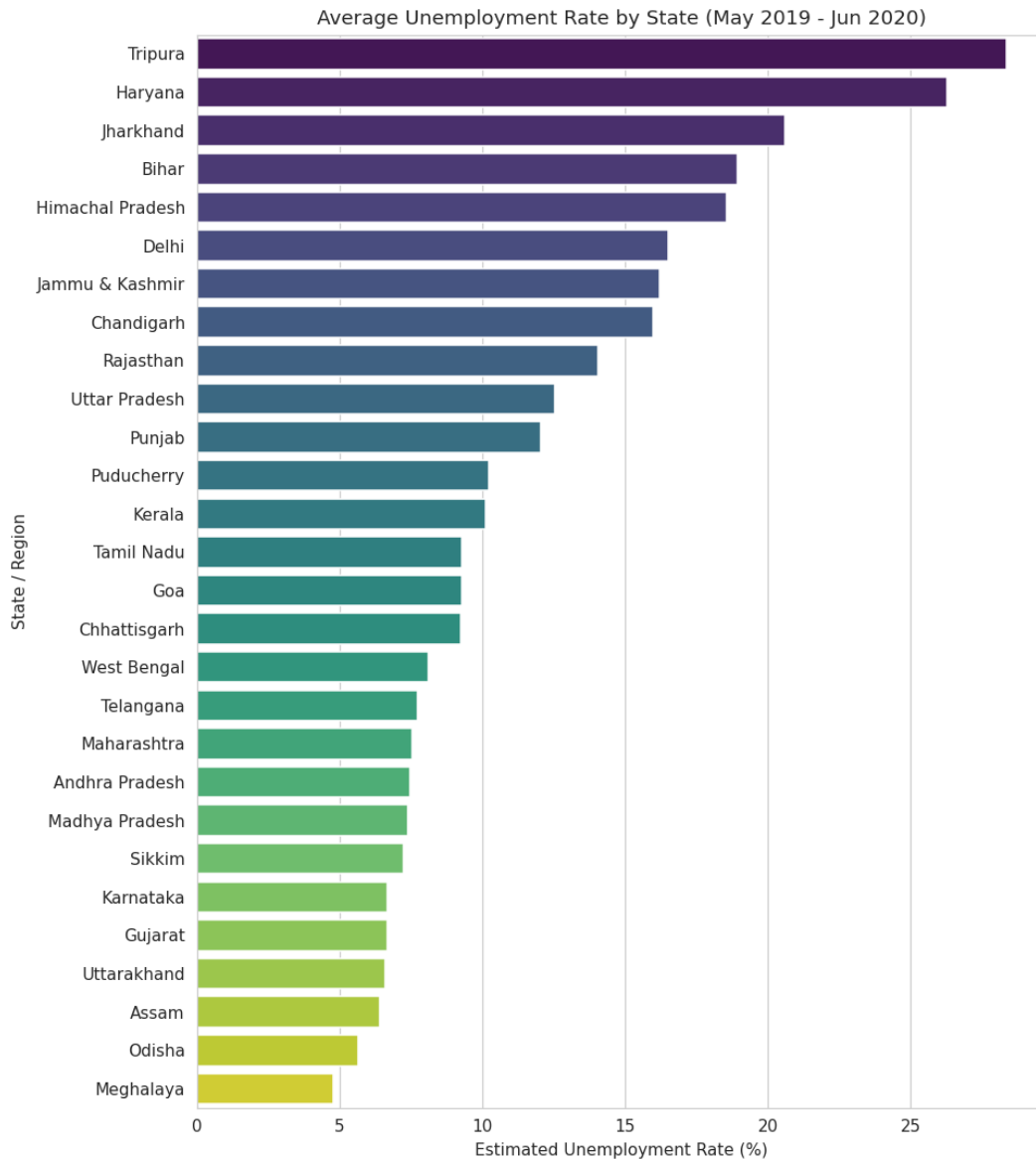


Figure 3. Average unemployment rate by state, May 2019 - June 2020.

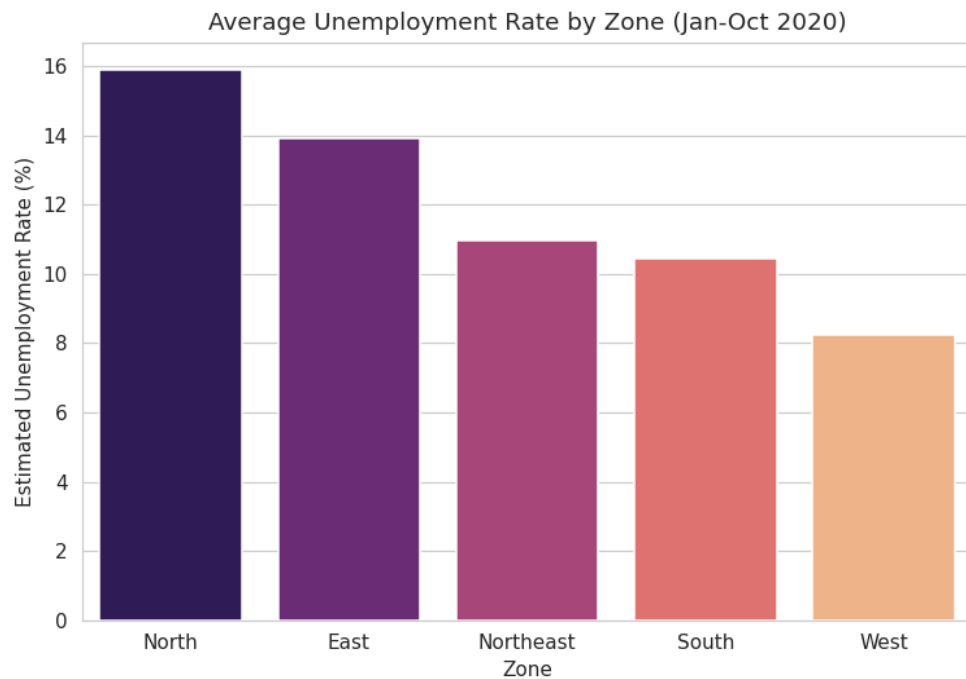


Figure 4. Average unemployment rate by geographic zone, Jan-Oct 2020.

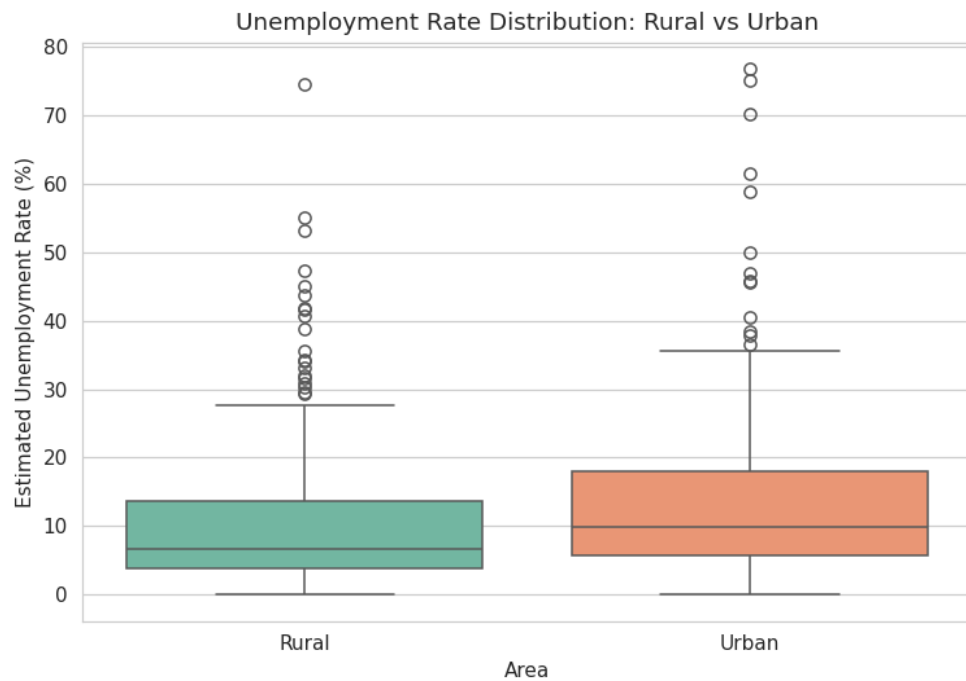


Figure 5. Unemployment rate distribution, rural vs. urban areas.

Distribution and Correlation

The unemployment rate distribution is right-skewed, with most observations clustered under 15% but a long tail of high-unemployment months driven by the COVID period. Estimated employed count and labour participation rate show only weak correlation with the unemployment rate itself, since they capture different aspects of the labor market (workforce size vs. joblessness rate).

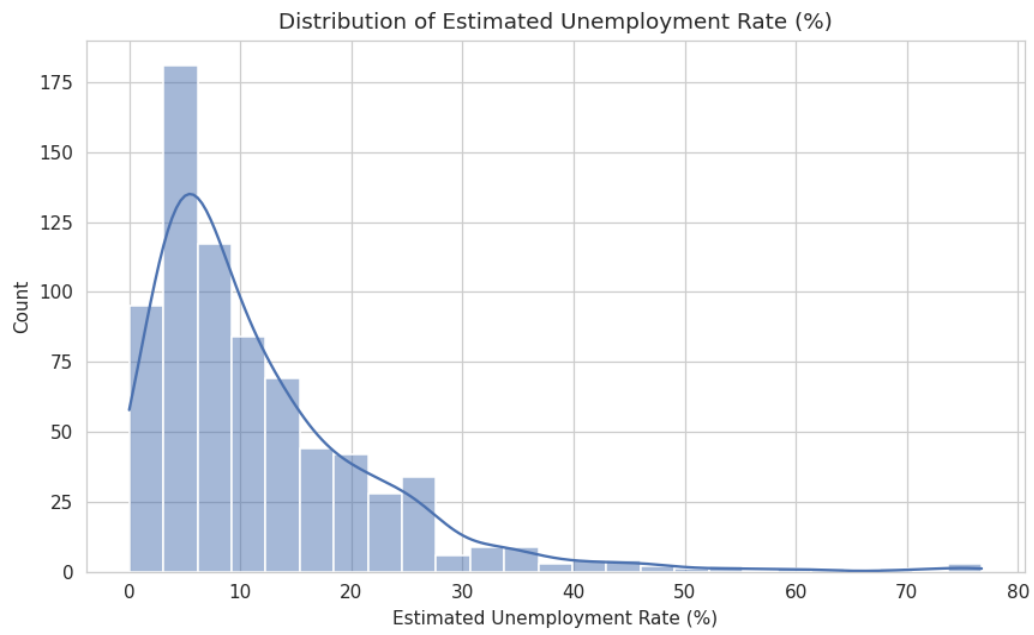


Figure 6. Distribution of the estimated unemployment rate.

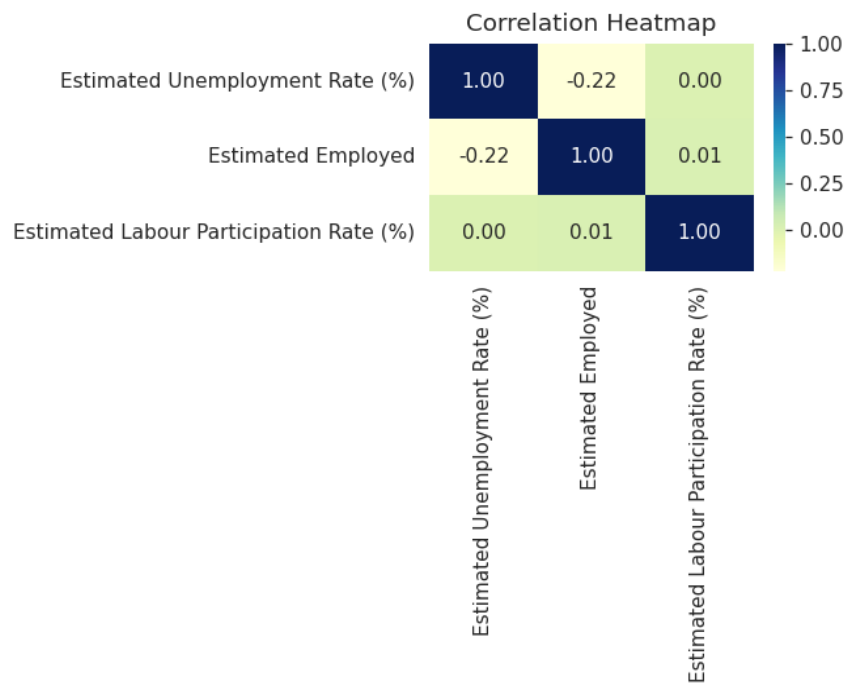


Figure 7. Correlation heatmap of numeric features.

Key Insights & Conclusions

1. National unemployment rose from 9.51% (pre-COVID) to 17.77% during the COVID period — an 8.26 percentage-point increase — peaking near 24.9% in May 2020.
2. Puducherry, Tamil Nadu, Jharkhand, and Bihar were the hardest-hit states, each seeing unemployment increases of 17-37 percentage points.
3. A handful of states (Jammu & Kashmir, Tripura, Himachal Pradesh, Chandigarh) actually saw unemployment ease slightly, suggesting the shock was not uniform nationwide.
4. The North zone carried the highest average unemployment through 2020, while the West zone remained comparatively resilient.
5. Labour participation fell alongside rising unemployment, indicating some workers exited the labor force entirely rather than remaining active job-seekers.

These findings support the well-documented narrative that India's COVID-19 lockdown produced an abrupt, severe, but geographically uneven labor-market shock, with the most acute effects concentrated in a handful of states during April-June 2020, followed by a partial recovery.